[In Progress] ICD-10-CM TITLES (US-ENG)

Overall Status: WORK IN PROGRESS

Jira ticket link: ☑ ENG-772513: [Frigate] R&D for the Replacement Diagnostic Classifications Entity (ICD-10) IN PROGRESS

Google Drive Link: Sign in to access Google Drive Folder

Supporting Research

Sources

- ICD-10 | CMS
- ICD-10-CM
- Medical Coding ICD-10-CM Chapters
- https://www.cms.gov/medicare/coding/icd10/downloads/032310_icd10_slides.pdf
- Structure of an ICD-10-CM Code | SEER Training
- https://www.healthnetworksolutions.net/index.php/understanding-the-icd-10-code-structure

Information

Scope of Research

- **Target Region:** United States (the CM, or Clinical Modification, is developed by CMS for US healthcare data).
- **Language:** English.
- **Data Characteristics:** The terminology consists of official medical conditions, administrative context phrases, and specific modifiers (such as laterality and encounter types).
- **Ambiguity:** ICD-10-CM codes are designed to map to single clinical concepts, making them generally less ambiguous by design. However, certain generated short terms (under 6 characters) were noted to concentrate acronyms and "trash".
- **Formatting/Affixing Notes:** The language relies heavily on hyphenations, apostrophes, and standard clinical abbreviations. Abbreviations in the official CMS source typically only occur when strictly necessary.

ICD-10-CM

The ICD-10-CM (International Classification of Diseases, Tenth Revision, Clinical Modification) is a standardized system used by healthcare providers to classify diagnoses and medical conditions for morbidity reporting. While it is derived from the World Health Organization's (WHO) core ICD-10 framework, the "Clinical Modification" is specifically tailored for the United States. Under authorization from the WHO, the National Center for Health Statistics (NCHS)—a division of the CDC—develops and maintains these U.S.-specific extensions. This ensures that while the codes meet domestic clinical needs, they remain strictly consistent with the global structures and conventions established by the WHO.

ICD-10-CM Code Structure

ICD-10-CM codes follow a hierarchical classification structure. Codes are organized into chapters, where each chapter corresponds to a specific body system, disease category, or external cause. For example, Chapter A00–B99 covers Certain Infectious and Parasitic Diseases.

Each ICD-10-CM entry contains several fields that describe the coded condition. These typically include an index identifier, the ICD-10-CM code, a flag value, a short description, and a long description.

For example:

```
00084  A081  0  Acute gastrent d/t Norwalk agent and oth small round vir
```

In this example, `A081` is the ICD-10-CM code. The short description provides a condensed form, while the long description provides the fully expanded medical terminology associated with the diagnosis.

ICD-10-CM Search Tool

The CDC provides an official search interface for exploring ICD-10-CM codes: https://icd10cmtool.cdc.gov/?fy=FY2026.

Term Enrichment - Integration of ICD-10-CM and External Medical Databases

Term enrichment is necessary because medical language is highly specialized and often expressed through multiple parallel vocabularies. The same medical concept may appear in formal scientific terminology, clinical abbreviations, or everyday language used by patients and healthcare staff.

Capturing these alternative expressions is essential for robust detection. For example, while "myocardial infarction" is the formal diagnostic term used in medical classification systems, the

same condition is commonly referred to as “heart attack” in clinical notes, patient communication, or informal documentation.

If detection relies solely on canonical ICD terminology, many valid references to medical conditions may be missed. In the context of Data Loss Prevention (DLP) systems, this limitation could allow sensitive Protected Health Information (PHI) to escape detection. Expanding the terminology set with synonymous expressions and lexical variants therefore improves recall and increases the overall effectiveness of PHI detection.

UMLS Integration

https://www.nlm.nih.gov/research/umls/index.html

The Unified Medical Language System (UMLS) integrates multiple biomedical vocabularies, including 74 English datasets and 71 datasets in other languages.

UMLS links synonymous terms using a shared Concept Unique Identifier (CUI). For each CUI, the API can return multiple atoms, where each atom represents a term variant from a specific source vocabulary.

Example:

Query Term	CUI	Source Vocabulary	AUI	Term String
cholera due to vibrio cholerae 01, biovar cholerae	C0494021	ICPC2ICD10DUT	A5068103	Vibrio cholerae 01; biovar cholerae

Each atom’s term string can be added as a lexical variant in the enriched terminology list.

CHV Integration

https://biomedinfo.smhs.gwu.edu/chv-files

The Consumer Health Vocabulary (CHV) focuses on expressions commonly used by healthcare consumers, including colloquial terms, ambiguous phrases, slang, and misspellings. During integration, most CHV terms were already present through UMLS or existing enrichment rules. Out of approximately 158,000 CHV terms, only 620 new variants were added after deduplication.

SNOMED CT Integration

https://www.nlm.nih.gov/healthit/snomedct/index.html

SNOMED CT is a comprehensive clinical terminology system used for electronic health information exchange. SNOMED CT data has been downloaded for analysis. Overlap with UMLS-derived terms will be evaluated to determine additional variants contributed by this source. *(Work in progress.)*

ICD-10-CM 2026 Dataset Description

The ICD-10-CM 2026 dataset contains 98,186 diagnosis entries, each represented as a textual description.

Word Count Distribution

At the token level, the dataset contains relatively short phrases, with an average of 9.66 words per entry and a standard deviation of 4.71 words. The shortest descriptions consist of a single word, while the longest extend up to 31 words.

The distribution indicates:

- 25% of entries contain 6 words or fewer
- The median is 9 words
- 75% of entries contain 13 words or fewer

Most ICD-10-CM descriptions are brief but sufficiently descriptive for clinical specificity.

Statistic	Value
Count	98,186
Mean	9.66
Std	4.71
Min	1
25%	6
50% (Median)	9
75%	13
Max	31

Text Preprocessing and Linguistic Analysis

To better identify the core semantic components of each diagnosis, the dataset was further analyzed using `spaCy` for word frequency and part-of-speech (POS) tagging.

From this analysis, two frequency-based vocabularies were constructed:

- Sign in to access Google Drive File : capturing high-frequency grammatical words
- Sign in to access Google Drive File : capturing common descriptive modifiers

Examples of high-frequency modifiers include:

- *unspecified, subsequent, other, right, initial, left, open, lower, nondisplaced, closed*

Examples of high-frequency function words include:

- *of, with, for, and, or, in, to, by, at, without*

Methodology of Building the ICD10CM Titles Database

Data Sources

- **CMS Official Files:** The foundational dataset consists of the latest 2026 ICD-10-CM files downloaded from the CMS website, including tabular order descriptions, code tables, index, addenda, and conversion tables .
- **External Vocabularies:**
  - **UMLS (Unified Medical Language System):** Leveraged via API to fetch term strings (atoms) and term variants based on Concept Unique Identifiers (CUIs).
  - **CHV (Consumer Health Vocabulary):** Integrated to capture everyday speech and informal medical variants.
  - **SNOMED CT:** Identified as a promising source for informal variants and term expansion.

Normalization and Enrichment Pipeline

Starting from an initial dataset of 98,186 leaf node entries , the terms were processed to generate an enriched list of over 335,000 entries.

- **Canonical Formatting:** Official terms were standardized into "Canonical terms" by converting to lowercase, trimming white spaces, and eliminating final punctuation .
- **Rule-Based Enrichment:** * *Simple Modifications:* Applied rules to remove or collapse hyphenation (e.g., non-hodgkin → non hodgkin / nonhodgkin), remove apostrophes, and replace "&" with "and".

- *Abbreviations:* Added common clinical abbreviations (e.g., syndrome → synd, chronic → chr, acute → acu, left/right → lt/rt) .
- *Synonym Expansion:* Standardized replaceable relational patterns, such as mapping "due to" to "because of" or "caused by".

- **Core Term Extraction:** To make matching more robust, an AI-assisted pipeline was utilized to identify the "core term" (the most essential part) of a code title—specifically targeting higher-level items with flag 0—and generating variations directly from that root.

Exploratory Analysis & Validation

- **PubMed Co-occurrence:** A Python script was used to fetch PubMed text snippets containing target phrases (e.g., "of cholera") to explore keyword co-occurrence patterns and typical medical constructions .
- **Bigram Analysis:** A frequency analysis of bigrams was run over the titles to surface recurring structural skeletons, such as relational phrases ("due to"), encounter types ("subsequent encounter"), and laterality ("of right").
- **Compositionality Script:** An `analyze_compositionality.py` script was developed to group terms into "pattern families," providing a structured way to craft regex patterns.
- **Sanity Checks:** Enriched entries were run against an ambiguity lexicon to catch false-positive-prone terms. The generated output was also cross-referenced with the EDK's current ICD-10-CM entity to evaluate coverage.
- **Manual Validation:** Datasets were segmented by length to facilitate manual review, allowing researchers to root out trash from short entries and independently validate long items.

AI-Powered Core Term Extraction

To enhance detection in data loss prevention systems, core terms (anchors) are extracted from ICD-10-CM titles. A core term represents the most essential concept within a diagnosis description. Once identified, these core terms can be expanded into multiple lexical variations to improve matching robustness across diverse expressions.

For example, given the description “Patient's noncompliance with other medical treatment and regimen due to unspecified reason,” the extracted core term would be `patient's noncompliance`. From this anchor, additional variations such as `pt noncompliance` and other common abbreviations or phrasing can be generated.

Implemented Core Terms Pipeline

The core terms extraction pipeline transforms the official ICD-10-CM FY2026 tabular file into a clean, DLP-ready term list for use as regex anchors. The pipeline is implemented as a series of

Python scripts in the [icd10cm-core-terms]() repository.

```
icd10cm_order_2026.txt
    + ai_extraction_checkpoint.jsonl
    + slot_vocabularies.json
         |
         ▼
extract_core_terms.py      →   core_terms.txt          (11,157 ICD-deri
         |
         ▼
expand_with_templates.py   →   core_terms.txt          (+ 6,514 templat
(uses slot_vocabularies.json + family_templates.json)
         |
         ▼
generate_variations.py     →   core_terms_variations.txt  (clean)
(uses slot_vocabularies.json)
         |
         ▼
mark_ambiguous.py          →   core_terms_variations.txt  (with * flags

```

Output Files

File	Description
`core_terms.txt`	17,671 cleaned core medical terms, one per line
`core_terms_variations.txt`	Same terms with all variations on one line, separated by `;`

Lines prefixed with `*` are flagged as ambiguous (high false-positive risk). 20 of 17,671 terms (0.1%) are flagged.

Step 1. Extract Core Terms

Script: `extract_core_terms.py`

Reads the fixed-width [ICD-10-CM order file]() and strips qualifiers, encounter types, laterality prefixes, severity/stage markers, and specificity suffixes. Filters out terms starting with digits, terms over 6 words, and single generic words.

After extraction, cross-references `ai_extraction_checkpoint.jsonl` (AI quality review): terms the AI rejected are dropped, terms the AI approved use the AI's canonical form, and AI-approved terms not reachable by the regex path are added as bonus coverage.

Elements stripped:

- Etiologic qualifiers: `due to`, `caused by`, `secondary to`, `following`, `complicating`
- Encounter types: `initial encounter`, `subsequent encounter`, `sequela`

- Laterality prefixes: `left/right/bilateral` (driven by `LATERALITY_TOKENS` from `slot_vocabularies.json`; e.g. `bilateral inguinal hernia` → `inguinal hernia`)
- Disease episode qualifiers: `in relapse`, `in remission`, `not having achieved remission`
- Severity/stage: `mild`, `moderate`, `severe`, `stage`, `grade`
- Specificity suffixes: `unspecified`, `NEC`, `NOS`
- Prefixes: `Other`, `Unspecified`, `Encounter for`

Step 2. Expand with Templates

Script: `expand_with_templates.py`

Uses `family_templates.json` (71 compositional families, e.g. `anatomy_x_condition`) and `slot_vocabularies.json` to generate anatomically-varied forms not present in ICD. For each condition token with ≥3 attested anatomy variants, generates the missing `condition of anatomy` combinations — producing clinically valid terms like `carcinoma of thyroid`, `tuberculosis of knee`, `thrombosis of aorta` that are absent from ICD under that exact form.

Parameter	Default	Effect
`MIN_ATTESTATIONS`	3	Minimum attested anatomy variants before expanding a condition
`SKIP_CONDITIONS`	see file	Conditions too generic to expand safely
`ADJECTIVAL_ANATOMY`	see file	Anatomy tokens that are adjectives, excluded from `of [anatomy]` position

Step 3. Generate Variations

Script: `generate_variations.py`

For each core term generates:

- **Plural forms** — rule-based, handles Latin/Greek irregular plurals (`calculus → calculi`, `diagnosis → diagnoses`)

- **Word-order reordering** — `"disorder of heart"` ↔ `"heart disorder"` — gated on `ANATOMY_TOKENS` from `slot_vocabularies.json`
- **Abbreviations & synonyms** — ~120 common conditions (`hypertension → HTN, high blood pressure`)
- **Anatomical substitutions** — `cardiac → heart`, `renal → kidney`, `hepatic → liver`, etc.
- **US/UK spelling variants** — `anemia ↔ anaemia`, `edema ↔ oedema`, `hemorrhage ↔ haemorrhage`, etc.
- **Bracketed abbreviations** from source — `[ECG]`, `[EKG]`, `[EOG]` extracted as separate variations

Step 4. Flag Ambiguous Terms

Script: `mark_ambiguous.py`

Prepends `*` to terms that are ambiguous in a DLP context.

Category	Examples
Environmental / external-cause codes	`bathroom`, `earthquake`, `tornado`, `motorcycle`
Generic clinical descriptors	`conditions`, `diseases`, `aftercare`, `onset`, `subacute`
Common words with non-medical uses	`stress`, `stroke`, `depression`, `shock`, `headache`
Medical terms that double as everyday words	`allergy`, `amnesia`, `jaundice`, `tremor`, `polyp`

Integration Files

File	Source	Purpose
`ai_extraction_checkpoint.jsonl`	AI review	Quality filter for extracted candidates
`slot_vocabularies.json`	Compositional analysis	Slot token lists (ANATOMY, CONDITION, LATERALITY, ...)

`family_templates.json`	Compositional analysis	71 structural template families for term expansion

ICD-10-CM Integrated Pipeline

The [ICD10cm-IntegratedPipeline]() creates a **bidirectional feedback loop** between compositional analysis and core term extraction to improve medical terminology vocabularies and template families.

Current Status

Metric	Value
Main Project Coverage	55.36% (218,028 / 393,844 terms)
Core Terms Generated	17,671 (11,157 ICD-derived + 6,514 template-expanded)
Anatomy Tokens Validated	141 (of 232 defined)
Production-Ready Templates	38 anatomy-bearing templates (of 71 total)
Conditions Suitable for Expansion	92 (with ≥3 attestations)

Top productive conditions: thrombosis (+122 terms), hernia (+121), ischemia (+121)

Key Goals

- **Vocabulary Validation** — Use empirical evidence from core terms to validate and refine slot vocabularies
- **Template Utilization** — Identify production-ready templates vs theoretical ones
- **Sync Management** — Maintain version consistency across projects
- **Curation Workflow** — Document and automate vocabulary improvement cycles

Implementation Roadmap

Phase	Description	Status
Phase 1	Documentation & Data Sharing: project structure setup, reference data, integration docs	In Progress

Phase 2	Vocabulary Validation: build validation script, generate reports, create refinement suggestions	Planned
Phase 3	Sync Infrastructure: build sync tool, add version metadata, optional git hooks	Planned
Phase 4	Template Analysis: build utilization analyzer, augment template metadata, generate reports	Planned
Phase 5	Curation Workflow: document workflow, build testing tools, create review checklist	Planned

Key Insights (from Empirical Validation)

- 141 noun anatomy tokens validated (vs 232 defined)
- 38 anatomy-bearing templates are production-ready (of 71 total)
- 92 conditions suitable for expansion (≥3 attestations)
- Top productive conditions: thrombosis (+122), hernia (+121), ischemia (+121)